RESEARCH SUMMARY

PAPER 1: *Closing Reasoning Gaps in Clinical Agents with Differential Reasoning Learning*

Problem and Motivation. The authors highlight that while large language models (LLMs) achieve high accuracy in factual question answering, clinicians need to trust the decisions they make. The chain-of-thought reasoning of LLMs often contains missing or hallucinatory steps, and existing evaluation metrics focus on final answers rather than the reasoning processes. This is dangerous because errors early in diagnostic reasoning can propagate to downstream decisions. Recent longitudinal benchmarks show that even state-of-the-art models achieve high final-diagnosis accuracy yet fail miserably at early differential diagnosis, underscoring the need for process-level supervision and auditability.

Method: Differential Reasoning Learning (DRL) converts a model's chain of thought into a directed acyclic graph whose nodes represent facts, hypotheses, and actions. The model's reasoning graph is compared with an expert reference using a clinically weighted graph edit distance to identify missing or hallucinated nodes and erroneous connections. Discrepancies are expressed as natural-language instructions and stored in a differential reasoning knowledge base (DR-KB), which is later used to retrieve relevant instructions and patch reasoning gaps during inference without changing the underlying model parameters.

Key Results: DRL was evaluated on multiple medical reasoning benchmarks, including MedQA, MedMCQA, and a Return Visit Admission dataset. It consistently improved accuracy relative to baseline LLMs and retrieval-augmented generation systems. Because the reasoning graph is explicit, clinicians can audit and understand each step, and new instructions can be added to the knowledge base to adapt to different clinical domains.

Important Highlights: DRL emphasizes process-level supervision, interpretability, and domain transfer. By focusing on intermediate reasoning rather than just final answers, it complements retrieval-augmented approaches like KARE and agentic medical knowledge graphs, which provide factual grounding but not reasoning quality.

PAPER 2: *What People See (and Miss) About Generative AI Risks*

Problem and Motivation: As generative AI systems become ubiquitous, there is growing concern about hallucinations, biased outputs, and other failures. The public's understanding of these risks is not well characterized, and policymakers lack data on which failure modes people recognize. Li et al. aim to understand public perceptions of generative AI risks by building on the U.S. National Institute of Standards and Technology (NIST) AI Risk Management Framework and focusing on failure modes rather than on specific applications.

Method: The authors created 24 realistic scenarios, each illustrating a different failure mode (e.g., hallucination, biased training data, privacy breach). After validation by responsible AI experts, the scenarios were presented to 960 U.S. participants. Respondents identified the failure mode, described the harm, and indicated whether we should address the issue.

Key Findings: Participants were highly aware of visible failures, such as hallucinations and offensive outputs, but upstream issues, such as data misuse or biased datasets, were often overlooked. Environmental impacts emerged as a concern. Most respondents believed responsibility is shared among multiple stakeholders, and those with greater trust in government were more supportive of strong AI regulation. Scenario-based questions elicited broader awareness than abstract surveys.

Important Highlights: The study suggests that AI literacy initiatives should also cover data collection and training practices. Scenario-based training could help clinicians to recognize a wider range of risks, and multi-stakeholder governance frameworks should address them.

GAP PROPOSAL AND VISION

Gap: Current clinical AI systems lack an integrated approach that models how a patient's condition evolves over time and simultaneously communicates the system's reasoning and associated risks. DRL and similar methods supervise reasoning on static cases, whereas retrieval-augmented frameworks such as KARE or AMG-RAG provide up-to-date information but do not capture temporal dynamics or support risk explanations. Meanwhile, studies on generative AI risk perceptions show that clinicians and patients often misunderstand upstream failures and misattribute responsibility. There is no framework that combines longitudinal reasoning, continuously updated knowledge, and risk-aware communication.

Why it matters: Accurate early reasoning is essential because errors in differential diagnosis can cascade through treatment plans. Without temporal reasoning, AI systems cannot update their conclusions when new data arrives. Transparent communication about model limitations and risks is vital for clinician and patient trust and for meeting regulatory requirements. Integrating reasoning and risk communication would reduce the risk of malpractice, improve safety, and foster adoption.

Business and clinical problem it solves: Unchecked generative-AI failures expose hospitals and practices to malpractice lawsuits, CMS penalties for inappropriate care, and wasted expenditures on unnecessary tests or readmissions. Clinically, the lack of risk-aware decision support means latent biases and data-collection flaws can propagate into diagnoses and treatment plans, eroding patient trust and leading to missed or delayed diagnoses. Integrating risk-aware generative AI education and reasoning into workflows directly addresses these liabilities by preventing harmful recommendations, improving care quality, and supporting regulatory compliance.

Proposed Approach: We propose Longitudinal Risk-Aware Differential Reasoning (LRDR), which extends DRL by linking reasoning graphs across time to reflect the patient's trajectory. New clinical observations update the graph, and a temporal attention mechanism revises hypotheses. A continuously updated medical knowledge graph supplies contextually relevant evidence. The system categorizes potential errors using the failure-mode taxonomy from the generative AI risk study and generates scenario-based explanations that communicate why the AI reached its conclusion and what risks remain. A feedback loop compares the evolving graph to clinical guidelines and updates the instruction database.

Evaluation plan: To validate the proposed approach, we will conduct a prospective study using the Return Visit Admission (RVA) longitudinal EHR dataset. Performance will be measured on early differential diagnosis (DDx) accuracy, calibration (e.g., Brier score), and the quality of risk communication assessed via clinician surveys. We will compare the full system against three baselines: (1) DRL without the temporal/risk-aware extensions; (2) a retrieval-augmented generation (RAG) baseline with no process supervision; and (3) a base LLM without any reasoning supervision or retrieval. Success will require at least a 10-percentage-point improvement in early DDx accuracy over the best baseline, along with statistically significant gains in calibration and risk-communication scores.

Expected Benefits: LRDR should improve differential diagnosis, reduce hallucinations through stronger grounding, and build trust through transparent, risk-aware explanations. Logging, reasoning, and risk assessments will facilitate regulatory compliance and streamline deployment. Ultimately, the framework aims to serve as a proof of concept for safer, more interpretable clinical AI.